



Multiclass

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Topics

- ▶ Multiclass prediction
 - ▶ **Decision Tree** example
- ▶ One vs. All
- ▶ Non-Parametric Models
- ▶ No free lunch theorem
- ▶ UCI machine learning repository



Handwriting Digit Recognition

0 1 2 3 4 5 6 7 8 9

0 1 2 3 4 5 6 7 8 9

Features:



Multiclass Prediction (example)

0 1 2 3 4 5 6 7 8 9

0 1 2 3 4 5 6 7 8 9

Features: 0 3 5 9



Feature Set

0	1	2	3	4	5	6	7	8	9
٠		٢	٣	٤	٥	٦	٧	٨	٩

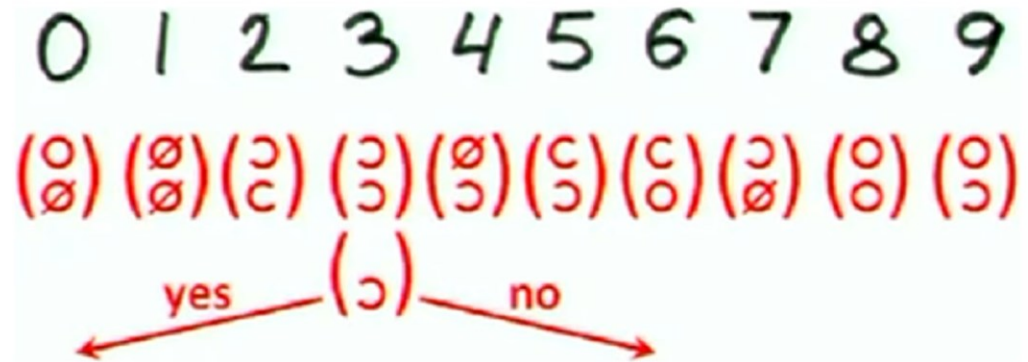


Feature Set (cont.)

0	1	2	3	4	5	6	7	8	9
$\begin{pmatrix} 0 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 2 \\ 2 \end{pmatrix}$	$\begin{pmatrix} 3 \\ 3 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 3 \end{pmatrix}$	$\begin{pmatrix} 5 \\ 5 \end{pmatrix}$	$\begin{pmatrix} 6 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 3 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 3 \end{pmatrix}$

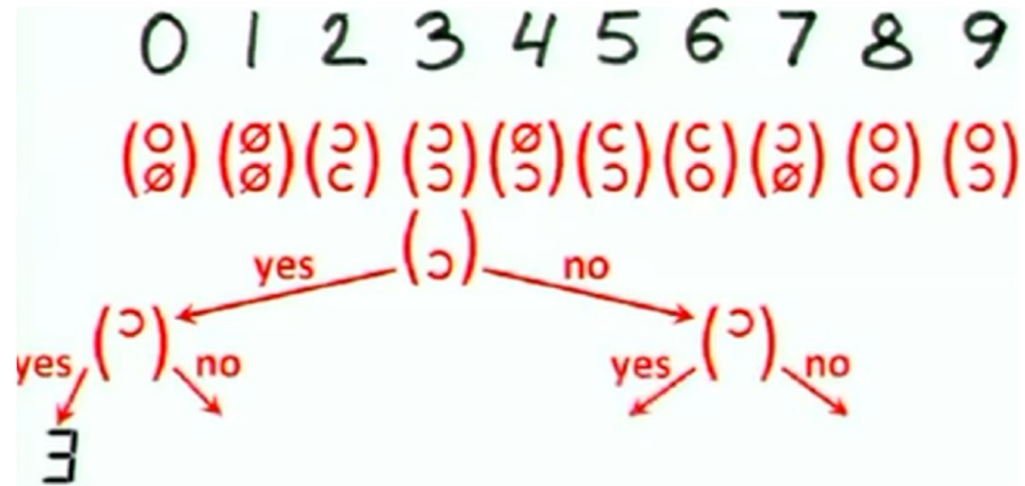


Decision Tree



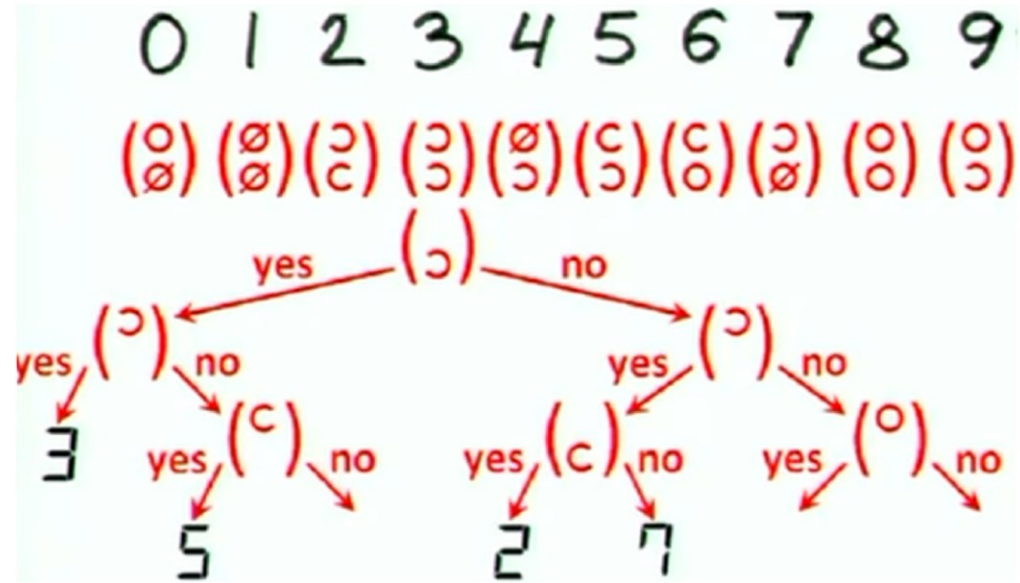


Decision Tree (cont.)



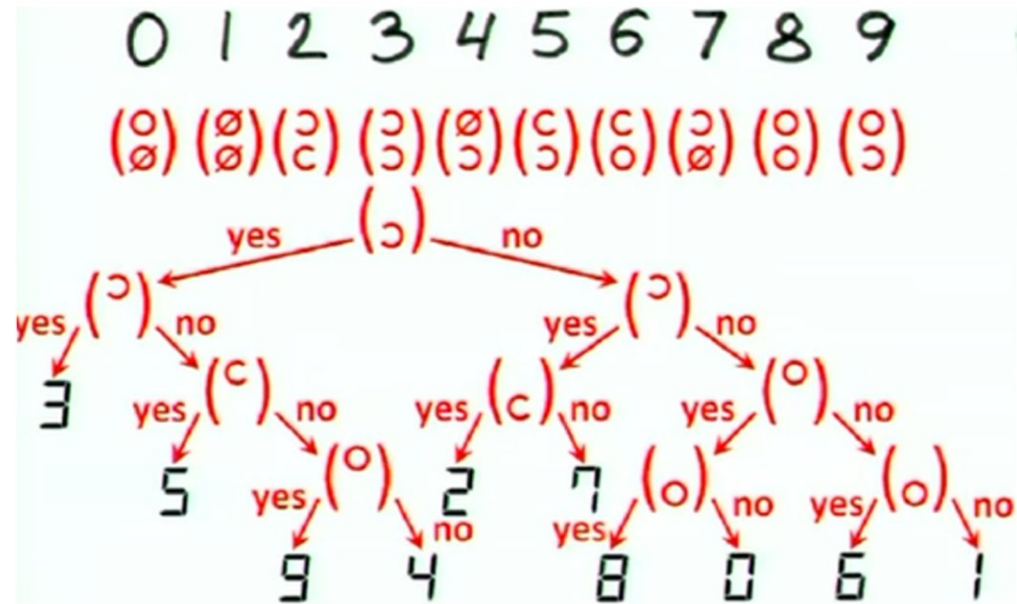


Decision Tree (cont.)





Decision Tree (cont.)





Confusion Matrix

► A multiclass (A, B, ..., E) prediction:

		Predicted				
		A	B	C	D	E
Actual	A					
	B					
	C					
	D					
	E					



Confusion Matrix (cont.)

► A multiclass prediction:

		Predicted				
		A	B	C	D	E
Actual	A	$\#TP_A$	$\#E_{AB}$	$\#E_{AC}$	$\#E_{AD}$	$\#E_{AE}$
	B	$\#E_{BA}$	$\#TP_B$	-	-	-
	C	$\#E_{CA}$	-	$\#TP_C$	-	-
	D	$\#E_{DA}$	-	-	$\#TP_D$	-
	E	$\#E_{EA}$	-	-	-	$\#TP_E$



Confusion Matrix (cont.)

► A multiclass prediction:

		Predicted				
		A	B	C	D	E
Actual	A	TP_A	E_{AB}	E_{AC}	E_{AD}	E_{AE}
	B	E_{BA}	TP_B	-	-	-
	C	E_{CA}	-	TP_C	-	-
	D	E_{DA}	-	-	TP_D	-
	E	E_{EA}	-	-	-	TP_E
		Accuracy				
		-	-	-	-	-



Performance Metrics

► How to Compute the followings (one vs. all):

► Accuracy

► True Positive (TP)

► False Negative (FN)

► False Positive (FP)

► True Negative (TN)

► Sensitivity

► Precision

...



Non-Parametric Models

- ▶ A non-parametric model
 - ▶ Does it refer to a model without parameters?
 - ▶ Examples:
 - ▶ K-nearest neighbors
 - ▶ Decision Tree
 - ...



Miscellaneous

- ▶ No free lunch (NFL) theorem
 - ▶ The theorem states that no learner can succeed on all learnable tasks - every learner has tasks on which it fails while other learners succeed [1].
- ▶ UCI machine learning repository
 - ▶ <https://archive.ics.uci.edu/ml/>



Further Reading

► **Understanding Machine Learning: From Theory to Algorithms, Shai Shalev-Shwartz and Shai Ben-David, Cambridge University Press, 2014.**

(Available online – No for distribution)